Ananya Vivek Kulkarni

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Education

2021 - Present **B.Tech, Dharmsinh Desai University** in Information Technology (CGPA: 8.68/10.0)

Work Experience

- **AI/ML Intern** — iHub-Data, IIIT Hyderabad *Jun'25 – Present*
  Supervisor: Mr. Govind Krishnan
  Working on designing and deploying deep learning models for adverse weather removal under the Mobility initiative at INAI, with a current focus on developing a lightweight model using knowledge distillation for real-time inference.
- **Research Intern - Space Applications Centre, ISRO**
  (Under Dr. Bipasha Paul Shukla, Dec '24 - Apr '25)
  Working on a Radiative Transfer Emulator for Greenhouse Gases, involving dataset creation and model building to emulate the behavior of gases like CO2, CH4, and NO2 for climate studies, while processing and generating over 9TB of data.
- **Research Intern - Defence Institute of Advanced Technology, DU, DRDO**
  (Under Dr. Amrita Nighojkar, May '24 - Jul '24)
  Developed AI-based models for leak detection in underground pipes for the maritime industry and authored a review paper on AI in maritime leak detection.

Skills Summary

Programming & Scripting:	Python, Java, C, C++, SQL, HTML, CSS, JavaScript, R, LaTeX
Frameworks & Libraries:	Django, React JS, Spring Boot, Spring MVC,Spring Data JPA, Flutter, Android Development, Kivy, KivyMd
Data Science & Analytics:	Pandas, NumPy, Matplotlib, Scikit-Learn, TensorFlow, PyTorch, NLTK, OpenCV
DevOps & Tools:	Git, GitHub, Docker, Kubernetes, Jenkins, CI/CD Pipelines
Databases:	MongoDB, PostgreSQL, MySQL
Soft Skills:	Leadership, Time Management, Teamwork, Problem Solving

Selected Technical Projects

- **Jigsaw Puzzle Solver Utilizing Self-Supervised Learning** *May '24 – Present*
  PyTorch, Pascal VOC, Self-Supervised Learning
  Developing a self-supervised model to solve jigsaw image puzzles. Generated a domain-agnostic dataset from Pascal VOC and trained contrastive embeddings to capture features, with a Streamlit UI currently in progress.
- **Car Detection in Low Visibility**
  (Hybrid Project-Image processing and Deep Learning Project, Jul '24 - Oct' 24)
  Utilized various image processing learning techniques such as CLAHE, DCP along with DEEPSORT and YOLOv8 for detecting and tracking cars in foggy weather in real time.
- **Automatic Information Tracker For Students**
  (Full-Stack Website based on Django Rest Framework, React JS, SQL, Dec '23 - Mar '24)
  Created a website to address the challenges faced by students when searching for suitable PG accommodation, hostel, and tiffin services in Nadiad.
- **Stress Relief Application**
  (Full-Stack Website based on Spring Boot, React JS, PostgreSQL, Docker, Jenkins, Dec'23 - Mar'24)
  Created a website with exercises to help with stress. Has the functionality of breathing exercises, gratitude journal writing, diary writing, etc. Containerized the application using *Docker* for seamless deployment and scalability. Implemented a *CI/CD pipeline* using *Jenkins* to automate builds and deployments.

Research Experience

- **SMART NOTE:::BOOK: Raspberry Pi-driven Affordable Image-to-Braille Converter for Visually Impaired Users**
  (IoT-based OCR project to convert text to Braille patterns, Feb '23 - Nov '23)
  – Developed an IoT-based text-to-Braille converter addressing the scarcity of Braille resources in developing nations.

- Mentored by *Assistant Prof. Viral H. Shah* and secured Rs.35,700 funding under the SSIP.
- Primary contributor & first author of **'Raspberryy Pi-Driven Affordable Image-to-Braille Converter for Visually Impaired Users'**.
- Received the **Best Paper Award** at *ASCIS 2023* and published in the **Springer CCIS series** (May 2024).

- **Career of Women Academicians in Engineering: A Case Study of Gujarat, India**
  Accepted at ICWSTCSC 2023, Dec '23
  Conducted a survey on the career trajectories of female academicians in engineering colleges in Gujarat, India, exploring factors contributing to their underrepresentation. This research was conducted under Associate Prof. Nikita P. Desai.
- **Development of Marathi Language Parser** *(Oct '23 - Sep '24)*
  Accepted at ERCICAM 2025, Presented on 8th March 2025
  Developed a Marathi language parser using Theory of Computation and Linguistics under Assistant Prof. Deepak Vegda, and authored the paper "Gender Agreement in Indo-Aryan Languages Using Rule-Based Parsing and Finite Automata" as the primary contributor and first author.
- **Car Detection in Low Visibilty** *Jul' 24 - Oct' 24*
  Accepted at OTCON 4.0, To be presented in April 2025
  Developed a Car Detection algorithm using various image processing techniques such as DCP, CLAHE along with deep learning techniques like YOLOv8 and DEEPSORT for tracking cars in foggy weather in real time.

Publications

1. **A. Kulkarni**, M. Shah, N. Thakur, S. Pednekar, and V. H. Shah, "Raspberry Pi-Driven Affordable Image-to-Braille Converter for Visually Impaired Users," in *Advancements in Smart Computing and Information Security*, S. Rajagopal, K. Popat, D. Meva, and S. Bajeja, Eds., Cham: Springer Nature Switzerland, 2024, pp. 225–242. ISBN: 978-3-031-59107-5.
2. S. Pednekar, **A. Kulkarni**, M. Shah, and N. Desai, "Women Academicians in Engineering Colleges in Gujarat, India: A Survey of the Current Scenario," Jan. 2024.

Patents

- **A System and Method for Real-Time Text-to-Braille Converter**
  Patent Filing Number: 202421066714, Filed: Sep 3, 2024
  Developed a real-time, cost-effective text-to-Braille converter designed for visually impaired students. The system features adjustable Braille output speed, voice activation, AI-powered image captioning, and a teacher monitoring tool for tracking student progress and storing data.

Honors, Awards, and Extracurriculars

- **Student Startup Innovation Policy (SSIP)** *(Government of Gujarat, Feb '23)*: Proposed the SMART NOTE:::BOOK project to address the scarcity of Braille books for visually impaired students.
- **Best Paper Award** at *ASCIS 2023 Conference (Dec '23)*.
- **Devheat Beta Hackathon**: Achieved Top 10 Finalist *(Jun '22)*.
- **Classical Dance (Bharatnatyam)**: Completed 3 years with Distinction.